Ananya Singh

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Experience

Data Engineer, Finix - San Francisco, CA Nov 2025 - Present

- Co-led a six-week migration of approximately 400 Redshift tables and reporting workflows to S3, Apache Iceberg, and Athena; built parity checks and phased the Superset cutover with **zero customer downtime**.
- Optimized Iceberg partitioning, table rewrites, and compaction, reducing data-platform spend approximately **40%** and critical pipeline runtime from 40 to **15-20** minutes; no dashboard pipeline failures observed after compaction.
- Built an Airflow customer-health data product covering approximately **2,000 applications** and **100 metrics** per customer segment; replaced inconsistent Finance queries with reconciled models used in monthly CEO and leadership reviews.
- Lead data engineering for a **Cube governance POC**, designing 5 fact, 4 dimension, and 5 aggregate models with YAML definitions and governed views. Partner with DevOps on AWS/EKS and Bedrock integration, assessing access boundaries and caching risks for PCI and PII data.
- Redesigned U.S./Canada bank and card-network reconciliation with Athena, Iceberg, Airflow, and Slack alerts; cut runtime from 2-3 hours to **30-40 minutes** and surfaced **five major incidents** involving missing wires or files.
- Built GitHub Actions workflows governing **all data-repository deployments**, replacing manual S3 uploads with approval checks, tracked test releases, conflict detection, and expiry cleanup; no deployment version-drift incidents reported after adoption.

Data Engineer, QR Associate, JPMorgan Chase - Plano, TX Feb 2024 - Nov 2025

- Led an event-driven risk-data migration consolidating **22 data panels into 9**, reducing processing time **40%** and data-quality issues **80%** for enterprise credit-risk forecasting.
- Built Python, Spark, and SQL validation tooling for quarterly model-attribute changes, eliminating **30+ hours** of manual work per cycle and accelerating data delivery **50%**.
- Engineered Great Expectations and DuckDB controls for C&I portfolio refreshes supporting CCAR, CECL, and Basel reporting, reducing downstream anomalies and schema issues **70%**.
- Enhanced distributed ETL and KPI models to close **$23B in credit-exposure gaps** and improve completeness, lineage, and monitoring across regulated risk datasets.

Quantitative Analyst Intern, CME Group - Chicago, IL May 2023 - Aug 2023

- Analyzed high-frequency market data with Python and SQL; engineered order-flow and mean-reversion features and evaluated predictive models for mid-price movement and execution efficiency.

Technical Skills

Languages & Processing: Python, SQL, PySpark, Apache Spark, Pandas, NumPy

Data Platform: Airflow, Iceberg, Athena, S3, Kafka / Amazon MSK, Debezium, Firehose, Glue Data Catalog, EMR Serverless, EKS, RDS PostgreSQL, Parquet

Modeling & Engineering: Cube, dimensional modeling, semantic layers, GitHub Actions, Terraform, Datadog, Superset, Great Expectations, DuckDB, CI/CD, CDC

Education

University of Illinois Urbana-Champaign May 2024
M.S. in Financial Engineering (Operations Research) | GPA: 3.97/4.00

BITS Pilani May 2021
B.E. in Chemical Engineering; Minor in Finance | GPA: 9/10